

BONITA: Generating Italian Instruction-Tuning Data for Boolean Question Answering

3-cfu Project Work in Natural Language Processing

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Abstract

High-quality instruction-tuning data is scarce for Italian, while unannotated Italian text is everywhere. Bonito (Nayak et al., 2024) showed that a model can learn to turn raw text into instruction-tuning data, writing tasks grounded in a passage together with their answers. We bring this idea to Italian with BONITA, a task generator for boolean question answering built on Minerva-7B, an LLM trained from scratch on Italian and English data. Given any Italian paragraph, BONITA writes a question about it and its answer, turning Italian Wikipedia into training data without any human annotation, and it produces well-formed tasks. Following Bonito, we evaluate BONITA through its data. We train a model only on BONITA's questions and compare it with the same model without this training, on a test set of human-labeled Italian questions. The student reaches 88.0 accuracy and 87.5 macro-F1 on average over three instruction wordings, against 80.1 and 79.9 for the baseline, and is better with each wording.

1 Introduction

Instruction tuning is the standard way to adapt a large language model (LLM) to a task, but it needs annotated instruction/response pairs, which are expensive to collect and scarce for languages other than English. Unannotated text, on the other hand, is abundant: the question is how to turn it into supervision.

Common answers to this problem have clear limits. Continued pre-training (next-word prediction) on in-domain text uses unannotated data directly, but does not teach the model to follow instructions and can even undo the benefits of instruction tuning (Nayak et al., 2024). Self-generation methods such as Self-Instruct (Wang et al., 2023) prompt a strong LLM to invent tasks, but the tasks are not grounded in a given text and depend on the quality of a large, often proprietary, model. Nayak et al. (2024) propose *conditional task generation*: Bonito, a 7B

model fine-tuned to read a passage and a task type and write a task grounded in that passage together with its answer. Running Bonito over unannotated text yields a synthetic instruction-tuning dataset, which they show to adapt LLMs to new domains better than self-supervision.

In this project we build **BONITA**, an Italian Bonito for a single task type, boolean question answering: from an Italian paragraph, BONITA writes a question about it and its answer (Vero/Falso). BONITA is a QLoRA (Dettmers et al., 2023) adapter on Minerva-7B-base (Orlando et al., 2024), an LLM trained from scratch on Italian and English data. Like Bonito, it starts from a pretrained model and is trained in Bonito's meta-template. Its training data are 80% of BoolQ-Italian, the Italian translation of BoolQ released by Sapienza NLP in ITA-Bench (Moroni et al., 2024).

As in Bonito, we evaluate BONITA by what its data is used for: (i) BONITA generates a dataset from 10,000 unannotated Italian Wikipedia paragraphs; (ii) a *student*, Minerva-7B-instruct with a new LoRA adapter, is trained on this synthetic data alone; (iii) the student and Minerva-7B-instruct without fine-tuning (the baseline) are scored by log-likelihood on a test set of 2,490 human-labeled BoolQ-Italian examples that neither BONITA nor the student has seen. To make the comparison robust to the wording of the prompt, the test set is scored three times, once for each of the instruction variants used in BONITA's training data.

Our main findings are:

- BONITA almost always produces well-formed tasks: 99.6% of 10,000 generations are kept after parsing and deduplication.
- A student trained only on BONITA's data beats the baseline with each of the three instructions, reaching 88.0 accuracy and 87.5 macro-F1 on average, against 80.1 and 79.9; its answers also depend less on the wording

of the instruction.

2 Background

Bonito. Nayak et al. (2024) remix existing prompted datasets, mostly from P3 (Sanh et al., 2022), into 1.65M *meta-template* examples, and use them to fine-tune the pretrained Mistral-7B into a conditional task generator. In a meta-template, the input contains a task type and an unannotated passage, delimited by special tokens (`<|tasktype|>`, `<|context|>`, `<|task|>`); the output is a task, i.e. an instruction that embeds the passage, followed by its solution after a `<|pipe|>` separator. To adapt a target LLM to a new domain, Bonito generates tasks from the domain’s unannotated text, and the target model is instruction-tuned on them. Across seven datasets covering yes-no QA, extractive QA and natural language inference, this improves Mistral-Instruct-v2 and instruction-tuned Mistral and Llama 2 by 22.1 F1 points on average over their zero-shot performance, while continued next-word prediction on the same text lowers it by 0.8 points. We reuse Bonito’s meta-template and its extrinsic evaluation protocol; our generator, training data, language and scale are different (Section 4).

BoolQ. BoolQ (Clark et al., 2019) is a reading-comprehension dataset of naturally occurring boolean (true/false) questions (from Google search queries), each paired with a Wikipedia paragraph that contains the answer. Answering often needs inference beyond lexical matching.

Minerva. Minerva (Orlando et al., 2024) is a family of LLMs pre-trained from scratch on Italian and English data, with an Italian-aware tokenizer. We use the 7B pretrained model (Minerva-7B-base) for BONITA, and its instruction-tuned version (Minerva-7B-instruct) for the student and the baseline.

Log-likelihood scoring. A multiple-choice task can be evaluated without parsing generated text: the model computes the log-likelihood of each candidate answer given the prompt, and the most likely one is its prediction. We use this protocol for the two answers `Vero` and `Falso`.

3 Data

BoolQ-Italian. We use BoolQ-Italian, a machine translation of BoolQ into Italian released by Sapienza NLP in ITA-Bench (Moroni et al., 2024) (links in Section 9), which keeps the original English passage in its metadata. It has only a `train` (9,399 rows) and a `validation` (3,259 rows) split, and 487 passages appear in both. We therefore merge the two splits (12,658 rows) and divide them 80/20 *by original passage*, so that all the questions about a paragraph end up on the same side: 10,094 rows are used to train BONITA and 2,564 are held out as the test set. BoolQ also contains near-duplicate paragraphs (e.g. two Wikipedia revisions of the same paragraph, with the same Italian translation) and repeated questions, which a split by exact passage does not separate. We remove the test examples whose Italian passage or normalized question (lower-cased, punctuation removed) also appears in BONITA’s data (74 examples), leaving **2,490 test examples**, 61.0% of which are labeled `Vero`.

Unannotated Italian text. BoolQ passages are Wikipedia paragraphs, so we take the unannotated text from the same source: the first of the ten Parquet shards of the Italian Wikipedia dump `wikimedia/wikipedia (20231101.it, ~180k articles)`. After shuffling the articles, we take one random paragraph from each. The paragraph must end with a full stop and be at least as long as the 5th percentile of BONITA’s training passages. In total we collect **10,000 paragraphs**, roughly the size of BoolQ’s training data. No label is attached to these paragraphs.

Dev sets. Both BONITA and the student keep a dev set of 5% of their training *passages* for model selection, so that no passage is shared between train and dev and the dev loss is not optimistic. No test example is used for training or model selection.

4 Approach

Our system follows Bonito (Nayak et al., 2024): a generator turns unannotated text into instruction-tuning data, and the data are evaluated extrinsically, by training a model on them and testing it on a human-labeled benchmark. In our case the task type is boolean question answering: from an Italian passage, BONITA writes a yes/no question about it and its answer, `Vero` or `Falso`. The pipeline, shown in Figure 1, has five stages:

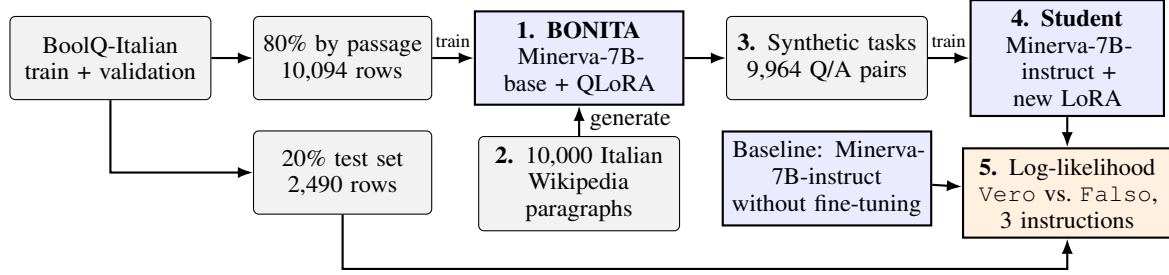


Figure 1: The BONITA pipeline. BONITA is trained on 80% of BoolQ-Italian (1), writes one question/answer pair for each unannotated Wikipedia paragraph (2–3), and a student is trained only on the synthetic tasks (4). The student and the baseline (the same 4-bit model, with the student’s adapter disabled) are scored on the test set, the remaining 20%, once for each of three instructions (5).

1. **Train BONITA:** a QLoRA adapter on Minerva-7B-base, plus the embeddings of its special tokens, trained on the 10,094 BoolQ-Italian rows of the training split.
2. **Collect unannotated text:** 10,000 paragraphs from Italian Wikipedia, the same kind of source as BoolQ, with no label attached.
3. **Generate a synthetic dataset:** BONITA writes one question/answer pair per paragraph; generations that cannot be parsed and duplicate questions are discarded.
4. **Train the student:** a new QLoRA adapter on Minerva-7B-instruct, trained only on the synthetic dataset, so that it learns the task only from data written by BONITA.
5. **Evaluate** the student and the baseline, Minerva-7B-instruct without fine-tuning, on the 2,490 test examples, by log-likelihood scoring of *Vero* and *Falso* with three instruction variants.

The following subsections describe the stages in order, and the last one separates what we take from Bonito from our own contribution.

4.1 Stage 1: training BONITA

Each of the 10,094 rows of BONITA’s data becomes a prompt/completion pair in Bonito’s meta-template:

Prompt: <pre>< tasktype > boolean question answering < context > {passage} < task ></pre> Completion: <pre>{instruction} {question} Vero o Falso? < pipe > Risposta: Vero/Falso</pre>

The prompt holds the task type and the passage.

The completion holds the task, that is the instruction followed by the question, and after `<|pipe|>` its answer, so BONITA learns to write both the question and its answer. BoolQ has no instructions, so we add one, sampled uniformly from three equivalent Italian wordings, e.g. “*Dopo aver letto il passaggio, rispondi alla seguente domanda:*” (all three in Appendix 10.1). With a single wording, BONITA would learn to start every task with the same sentence; with three, the instruction varies across the synthetic tasks.

Special tokens. The four markers are added to the tokenizer as special tokens, so that each one is a single, unsplittable token. Minerva-7B-base has no free embedding rows, so the input embeddings and the output layer are resized. The four new rows are trained together with the LoRA weights through PEFT’s `trainable_token_indices`, in both the input embeddings and the output layer, while the rest of the embeddings stays frozen.

4.2 Stages 2–3: generating the synthetic dataset

The unannotated paragraphs are collected as described in Section 3. For each Wikipedia paragraph, BONITA receives the prompt part of the meta-template and generates the task and its answer with greedy decoding (for reproducibility), at most 256 new tokens, stopping as soon as `Risposta: Vero` or `Risposta: Falso` is produced. Prompts are batched (16 per batch) with left padding, so that every prompt ends right where generation starts. Minerva’s configuration disables the key-value cache, so we enable it explicitly for generation. The `<|pipe|>` token is removed when decoding, and a generation is parsed with a regular expression on

Risposta: Vero/Falso; everything before the match is the task (instruction + question). It is kept only if the answer is parseable, the task ends with “*Vero o Falso?*”, and its normalized question is not a duplicate of an already kept one.

4.3 Stage 4: training the student

The student is a new LoRA on Minerva-7B-instruct, trained only on the synthetic tasks, each formatted as in evaluation:

```
{passage}

{instruction} {question} Vero o Falso?
Risposta:
```

with completion “Vero” or “Falso”, each with a leading space. The task in the prompt is the one written by BONITA, instruction included. Prompts are given as plain text, without Minerva-7B-instruct’s chat template, both in training and in evaluation.

4.4 Stage 5: log-likelihood evaluation

Each test example is formatted with the student prompt above. The model computes $\log P(c \mid \text{prompt})$ for $c \in \{\text{“Falso”}, \text{“Vero”}\}$, summing the log-probabilities of the continuation tokens, and predicts the most likely one. A continuation can span several tokens (“Vero” is one token, “Falso” two), and each token after the first is predicted given the previous ones, so each (prompt, continuation) pair is scored in its own forward pass. Prompt and continuation are tokenized separately and concatenated, so that the prompt tokens are identical for both candidates; requests are right-padded and batched, and the log-probability of the token at position t is read from the logits at position $t - 1$. This removes any dependence on output formatting and answer parsing.

The test set is scored three times, once for each of the three instructions, always the same for all the examples and both models; we report the results for each instruction and their mean. This makes the comparison less dependent on a single wording of the prompt.

A controlled comparison. Minerva-7B-instruct is loaded once in 4-bit with the student’s adapter: the student is scored with the adapter on, the baseline with the adapter turned off, on identical prompts. Any difference in scores is therefore due to fine-tuning on the synthetic data.

Generations	10,000
malformed (no answer / suffix)	4
duplicate question	32
Kept tasks	9,964 (99.6%)
Share of Vero labels	71.3%

Table 1: Synthetic dataset generated by BONITA from 10,000 Italian Wikipedia paragraphs.

4.5 Original contribution

The meta-template, the overall idea of conditional task generation and the extrinsic evaluation protocol are taken from [Nayak et al. \(2024\)](#). Our contributions are the Italian instantiation of Bonito on Minerva with QLoRA, the parsing and filtering of the generations, the evaluation with multiple instructions, and the whole training, generation and evaluation code.

5 Experiments

Training setup. All models are loaded in 4-bit NF4 with double quantization and `bfloat16` compute, with a LoRA adapter ([Hu et al., 2022](#)) on all linear projections ($r = 16$, $\alpha = 32$, dropout 0.05; $\sim 42\text{M}$ trainable parameters, 0.56% of the model), i.e. QLoRA ([Detmeters et al., 2023](#)). Before adding the adapter, the base weights are frozen, layer norms are cast to `float32`, and gradient checkpointing is enabled to fit a 7B model with batch size 8 on a single 24 GB GPU.

Both BONITA and the student are trained with supervised fine-tuning (`trl`’s `SFTTrainer`) on prompt/completion pairs, with the loss computed on the completion only. They share the same hyperparameters: effective batch size 16 (8×2 accumulation steps), AdamW with learning rate 5×10^{-5} , cosine decay and 3% warm-up, at most 4 epochs. The dev loss is evaluated every 100 steps, the checkpoint with the lowest dev loss is restored at the end, and training stops early after 3 evaluations without improvement. BONITA is trained on 9,611 rows with a dev set of 483, the student on 9,466 rows with a dev set of 498.

Generation. Table 1 reports the outcome of the generation stage. Only 4 of 10,000 generations cannot be parsed and 32 repeat an already kept question, so 9,964 tasks (99.6%) are kept for the student.

Metrics. Since the test classes are unbalanced (61.0% Vero), we report accuracy and macro-F1,

Model	Instr.	Accuracy	Macro-F1
Always Vero	–	61.0	37.9
Baseline	0	79.7	79.4
	1	81.1	80.9
	2	79.6	79.4
	mean	80.1	79.9
Student	0	88.2	87.6
	1	87.7	87.3
	2	88.2	87.7
	mean	88.0	87.5

Table 2: Results on the 2,490 BoolQ-Italian test examples (log-likelihood scoring), for each of the three instruction variants and their mean. Baseline: Minerva-7B-instruct without fine-tuning.

together with the majority baseline that always answers Vero.

6 Discussion

Quantitative analysis. Table 2 reports the results on the test set for each instruction and their mean. Both models are well above the majority baseline. The student is better than the baseline with each of the three instructions, and on average by 7.9 accuracy and 7.6 macro-F1 points. Its results also vary less with the instruction: its accuracy ranges from 87.7 to 88.2, the baseline’s from 79.6 to 81.1.

We then look at where the gain of the student comes from.

The gain comes from the Vero examples. Averaged over the instructions, the recall on Vero rises from 74.6 for the baseline to 88.2 for the student, while the recall on Falso barely changes, from 88.8 to 87.7 (Table 3). The baseline answers Vero on only 49.9% of the test examples, against 61.0% in the test set. The student answers Vero on 58.6%: more often than the baseline, but well below the 71.3% of its training data.

The gain is not only a matter of answering Vero more often. Every example has a margin, $\log P(\text{“Vero”}) - \log P(\text{“Falso”})$, positive when the model answers Vero. We move the decision threshold on the margin to the value that maximizes the baseline’s accuracy on the test set. Since the threshold is chosen on the test set itself, the result is an optimistic upper bound. Even so, the baseline only rises from 80.1 to 82.4, which accounts for about 2 of the 8 points gained by the student. With the same procedure, the accuracy of the student barely changes, from 88.0 to 88.3.

The student is more robust to the wording. Its

Model	Instr.	Pred. Vero	Recall Vero	Recall Falso
Baseline	0	50.8	75.0	87.0
	1	50.8	76.2	88.8
	2	48.0	72.7	90.4
	mean	49.9	74.6	88.8
Student	0	59.2	88.9	87.0
	1	57.9	87.4	88.2
	2	58.5	88.3	88.0
	mean	58.6	88.2	87.7

Table 3: Share of test examples answered Vero and recall of each class. 61.0% of the test examples are labeled Vero.

Baseline	Student			total
	wrong for all	right for some	right for all	
wrong for all	176	25	198	399
right for some	33	9	159	201
right for all	56	34	1,800	1,890
total	265	68	2,157	2,490

Table 4: Test examples by the number of instructions with which each model answers correctly: all three, none, or only some of them.

answer is the same with the three instructions on 97.3% of the test examples, against 91.9% for the baseline. Of the 201 examples that the baseline answers correctly only with some instructions, the student answers 159 correctly with all three (Table 4).

Error analysis. We consider the test examples that each model answers in the same way with all three instructions. The student fixes 198 of them that the baseline gets wrong (examples in Appendix 10.3), but also gets wrong 56 that the baseline answers correctly (Table 4).

Some errors come from the test set rather than from the models. First, some gold labels are wrong: “Il carbonio è un metallo nel sistema periodico?” is labeled Vero, while its passage says that carbon is a non-metallic element, and both models answer Falso. Second, the machine translation sometimes changes the meaning of the question, so that the gold label, which comes from the English question, no longer fits the Italian one. For instance, “Is a lightning bug the same as a firefly?” becomes “La lucciola è la stessa cosa di una lanterna volante?”, a paper sky lantern. Examples of this kind appear both among the errors of the two models and among the examples that the student fixes, where the student is counted as right only because the gold label does not fit the Italian question (Appendix 10.2).

7 Conclusion

We built BONITA, an Italian Bonito for boolean question answering, and evaluated it extrinsically. From 10,000 unannotated Wikipedia paragraphs BONITA writes 9,964 tasks, and a student trained only on them reaches 88.0 accuracy on the test set, against 80.1 for Minerva-7B-instruct, without any human annotation. The gain is not only a shift towards answering *vero* more often, and the student is more robust to the wording of the instruction.

The main limitations are the following. BONITA learned from BoolQ-Italian how to write its questions and applies them to Wikipedia, the same source as BoolQ, so the results do not tell how well the pipeline works on a different domain. Without a model trained with next-word prediction on the same paragraphs and a student trained on the human labels, we cannot tell how much of the gain comes from BONITA’s questions and labels rather than from the extra text or the answer format. BONITA covers a single task type, and some test examples have wrong labels or translations. Future work could apply BONITA to a specialized Italian domain, add the two missing comparisons, train BONITA on more task types, and clean the test set.

8 AI Report

Claude Code (Anthropic, model Claude Opus 5.5) was used to correct the report and to help write the code of the pipeline. All its outputs were reviewed and edited by the authors, who are responsible for the content of the report and of the code.

References

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9 Links to external resources

- Code: <https://github.com/francescobaio/bonita-it-boolq>
- Bonito (original code and models): <https://github.com/BatsResearch/bonito>
- BoolQ-Italian: https://huggingface.co/datasets/sapienzanlp/boolq_italian
- BONITA’s base model: <https://huggingface.co/sapienzanlp/Minerva-7B-base-v1.0>
- Student’s base model: <https://huggingface.co/sapienzanlp/Minerva-7B-instruct-v1.0>
- Italian Wikipedia: <https://huggingface.co/datasets/wikimedia/wikipedia> (20231101.it)

10 Appendix

10.1 Instruction variants

The three instructions added to BoolQ in BONITA’s training data and used to score the test set, numbered as in Tables 2 and 3.

0. *Dopo aver letto il passaggio, rispondi alla seguente domanda:*
1. *In base al testo precedente, rispondi alla domanda:*
2. *Secondo quanto riportato nel passaggio, rispondi alla domanda:*

10.2 Translation errors in the test set

In these test examples the Italian translation changes the question, so that the gold label, which comes from the English question, does not fit the Italian one. Both models answer `Falso` to the last three, and are counted as wrong; the student answers `Vero` to the first two, and is counted as right.

English question, gold	Italian translation
Vero	
Is a lightning bug the same as a firefly?	La lucciola è la stessa cosa di una lanterna volante ?
Is a chrysanthemum the same as a mum?	Un crisantemo è la stessa cosa di una margherita ?
Is a lightning bug and a firefly the same?	La lucciola e la lucertola sono la stessa cosa?
Are the medieval and middle ages the same?	Il Medioevo e l' età moderna sono la stessa cosa?
Are crows and ravens the same type of bird?	I corvi e i corvi marini sono la stessa specie di uccello?

Table 5: Test questions whose translation turns a pair of synonyms into two different things.

10.3 Examples fixed by the student

Test examples that the baseline gets wrong and the student answers correctly with all three instructions, chosen among those without translation or label problems. Each shows the Italian question, its gold label and an excerpt of the passage.

- *C'è un casinò a Catalina Island?* — `Falso`. Passage: “Il Catalina Casino deve il suo nome al termine italiano ‘casino’, che significa ‘luogo di ritrovo’. Nella struttura non è possibile giocare d’azzardo.”
- *L’ALS è la stessa malattia di Lou Gehrig?* — `Vero`. Passage: “La sclerosi laterale amiotrofica (SLA), nota anche come malattia dei motoneuroni (MND) o morbo di Lou Gehrig, è una malattia specifica [...]”
- *La Polonia faceva parte della Società delle Nazioni?* — `Vero`. Passage: “I membri [...] in questo periodo erano Argentina, Australia, [...] Perù, Polonia, Portogallo, [...]”
- *Le regole di Robert per il regolamento consentono il voto per delega?* — `Falso`. Passage: “Il voto per delega è automaticamente vietato nelle organizzazioni che hanno adottato il Robert’s Rules of Order Newly Revised [...]”
- *La Francia dichiarò guerra alla Germania nella seconda guerra mondiale?* — `Vero`. Passage: “La dichiarazione di guerra da parte della Francia

e del Regno Unito fu data il 3 settembre 1939, dopo che le forze tedesche invasero la Polonia.”